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(54) **ROTOR BLADE SYSTEM FOR TURBINE ENGINES**

(71) Applicant: **General Electric Company**,
Schenectady, NY (US)

(72) Inventors: **Keegan Nathaniel Mehrstens**, Peabody,
MA (US); **Yoon Seok Choi**, Kenwood,
OH (US); **Ramakrishna Mallina**,
Mason, OH (US); **Robert Bruce**
Dickman, Cincinnati, OH (US)

(73) Assignee: **General Electric Company**, Evendale,
OH (US)

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2240/307 (2013.01); **F05D 2250/25** (2013.01)

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2240/304
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(56) **References Cited**

U.S. PATENT DOCUMENTS

2,582,107 A 1/1952 Dakin
5,031,313 A * 7/1991 Blair B23P 6/002
29/402.06

5,167,489 A * 12/1992 Wadia F02K 3/04
415/218.1
5,642,985 A * 7/1997 Spear F01D 5/141
415/181
6,428,278 B1 8/2002 Montgomery et al.
7,794,202 B2 * 9/2010 Borthwick F01D 5/142
416/243
8,967,974 B2 3/2015 Garcia-Crespo
9,097,125 B2 8/2015 Ghorbani Zarimahalleh et al.
(Continued)

FOREIGN PATENT DOCUMENTS

EP 3075955 A1 10/2016
EP 3208467 B1 3/2021

OTHER PUBLICATIONS

Choi et al., "Analysis of Structural Mistuning Effects on Bladed
Disc Vibrations Including Aerodynamic Damping," International
Compressor Engineering Conference, School of Mechanical Engi-
neering, Paper 1627 (2004).

(Continued)

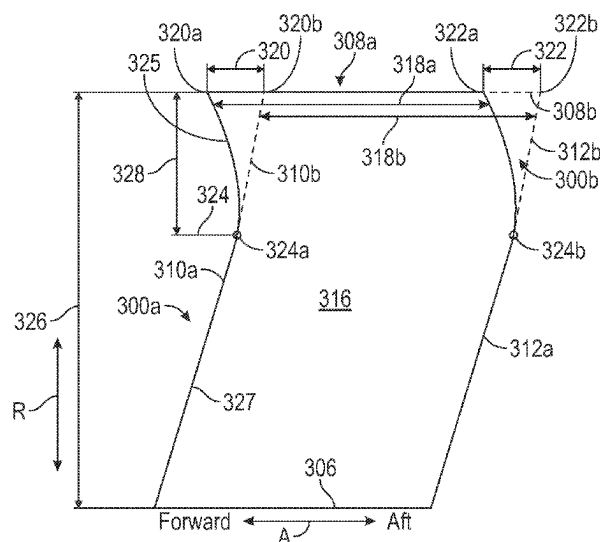
Primary Examiner — Michael L Sehn

(74) *Attorney, Agent, or Firm* — McCarter & English,
LLP

(57) **ABSTRACT**

A rotor blade system for a turbine engine having a longitu-
dinal centerline axis. The rotor blade system includes a rotor
configured to rotate about the longitudinal centerline axis
and a plurality of rotor blades coupled circumferentially
around the rotor, each rotor blade having a leading edge, a
trailing edge, a suction side, and a pressure side. At least one
rotor blade of the plurality of rotor blades has an axial sweep
of both the leading edge and the trailing edge as compared
to a base blade of the plurality of rotor blades.

18 Claims, 7 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

9,429,026	B2	8/2016	Schreiber
10,584,591	B2	3/2020	Opoka et al.
10,808,541	B2	10/2020	Bales et al.
10,837,459	B2	11/2020	Veitch et al.
10,851,655	B2	12/2020	Gonzalez-Gutierrez
11,002,293	B2	5/2021	Anand et al.
11,220,913	B2	1/2022	Molnar, Jr.
11,371,357	B2	6/2022	De Gaillard et al.

OTHER PUBLICATIONS

Ren et al., U.S. Appl. No. 17/661,796, filed May 3, 2022.

Andreoli et al., U.S. Appl. No. 17/649,030, filed Jan. 26, 2022.

* cited by examiner

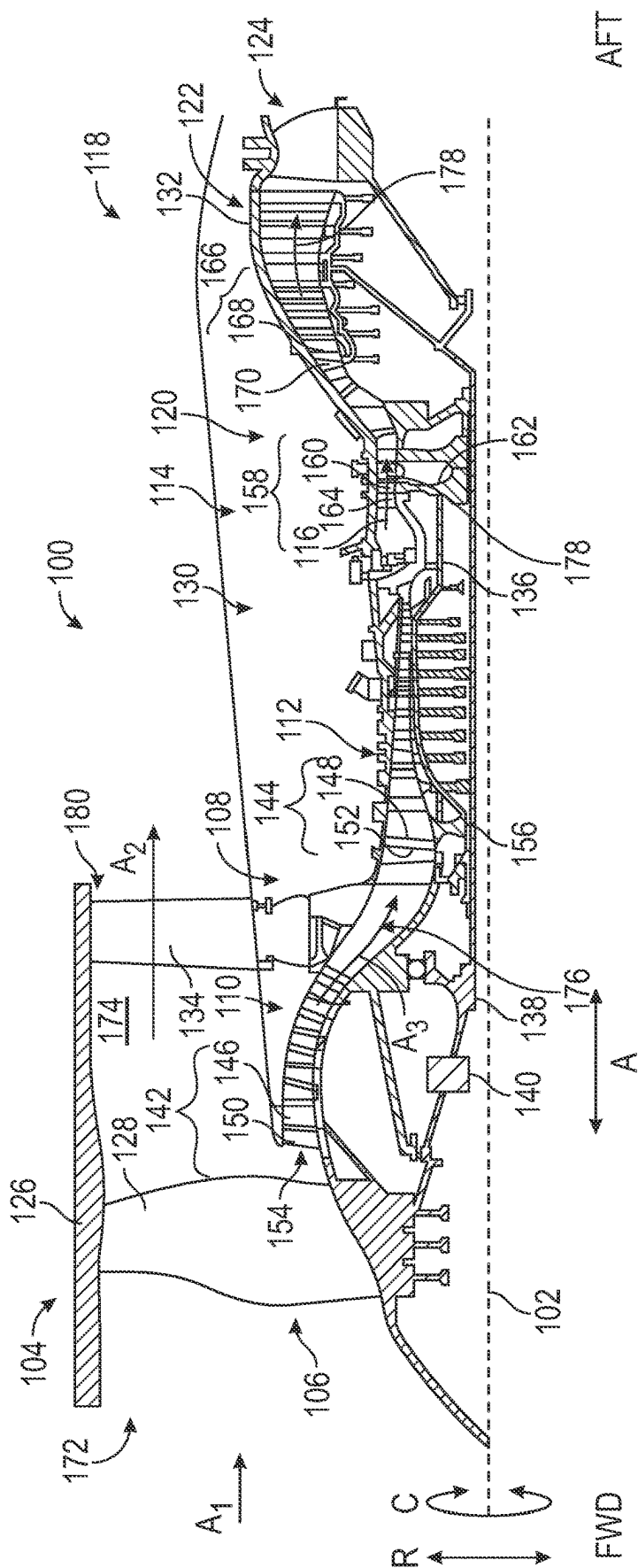


FIG. 1

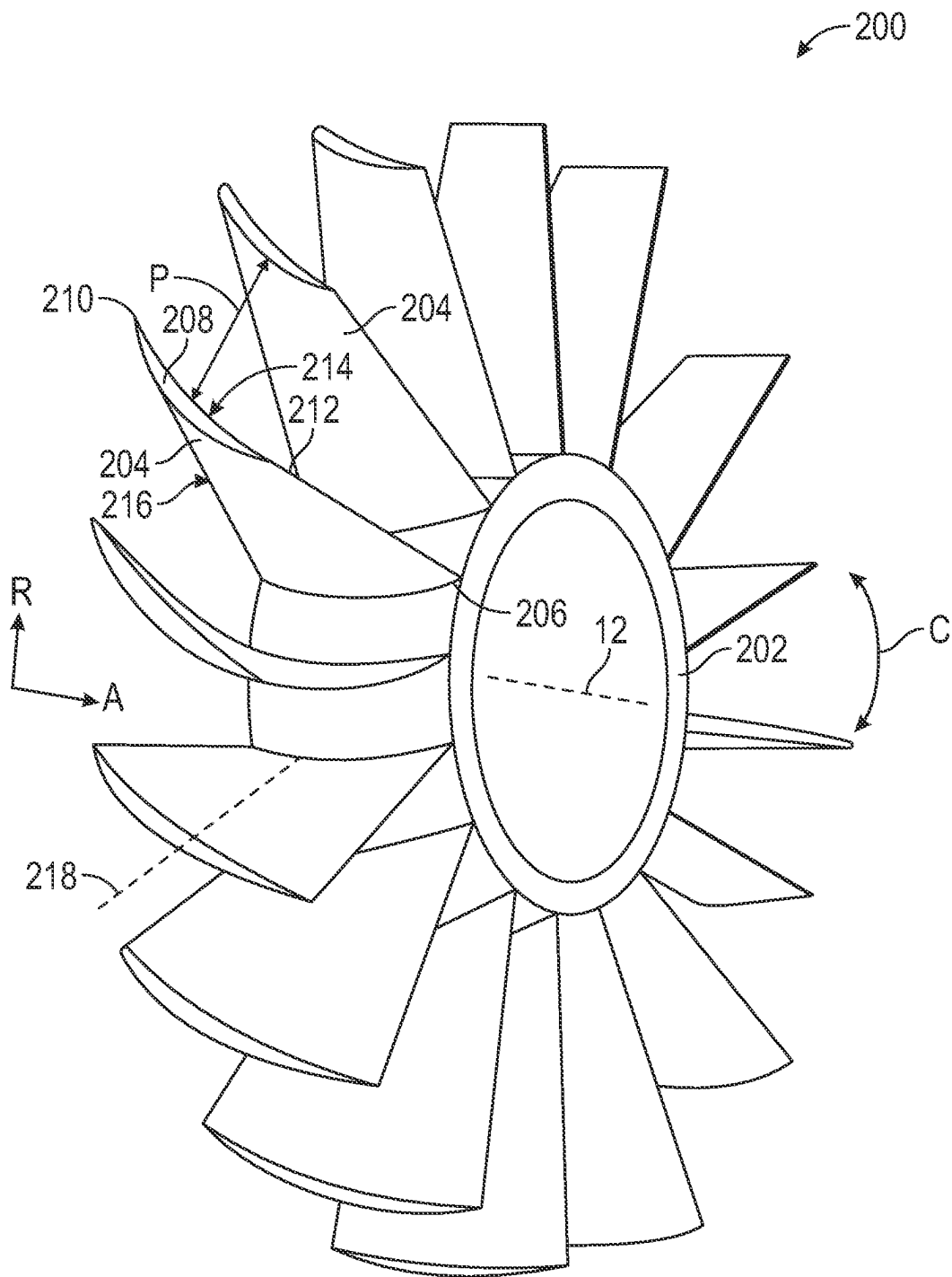


FIG. 2

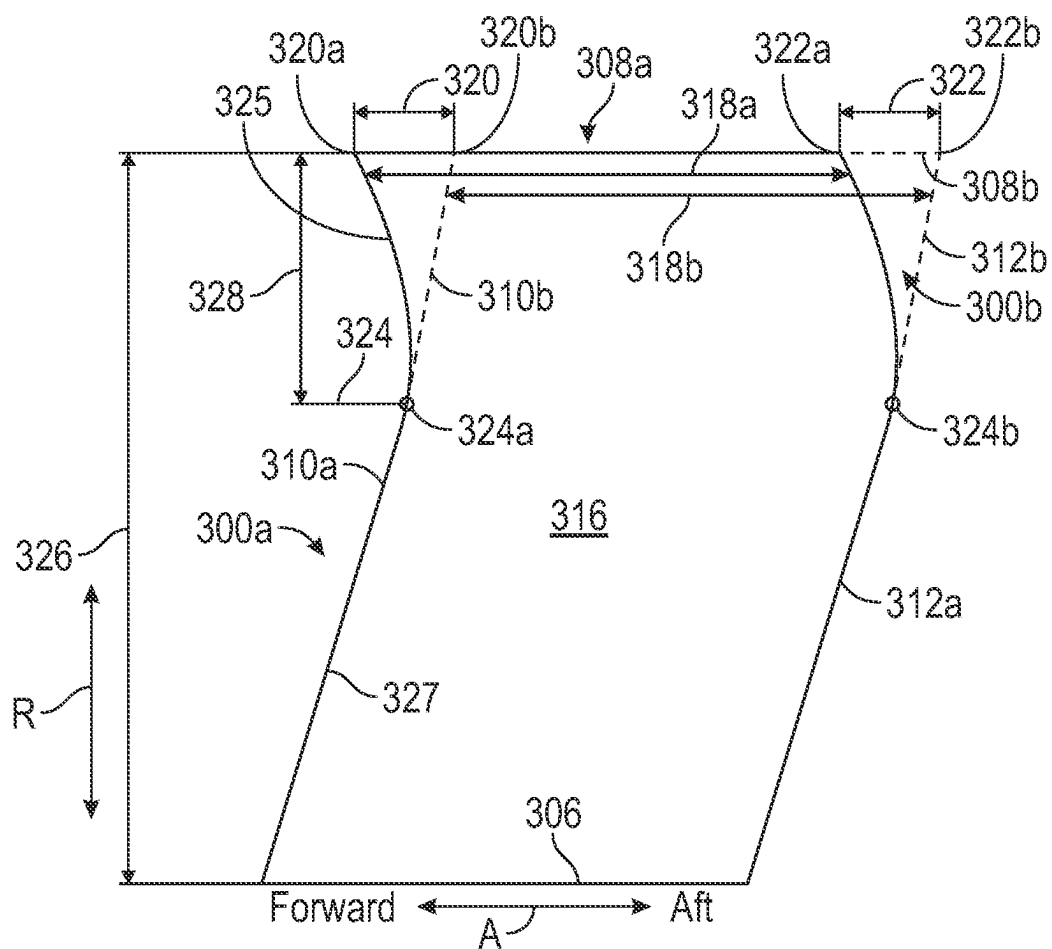


FIG. 3A

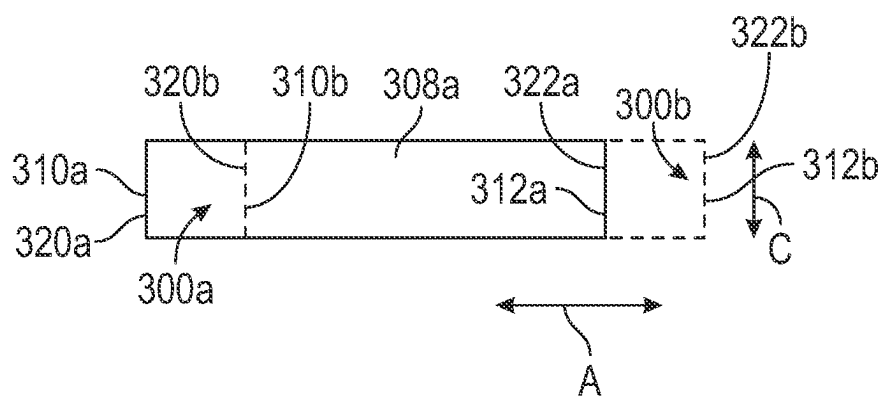


FIG. 3B

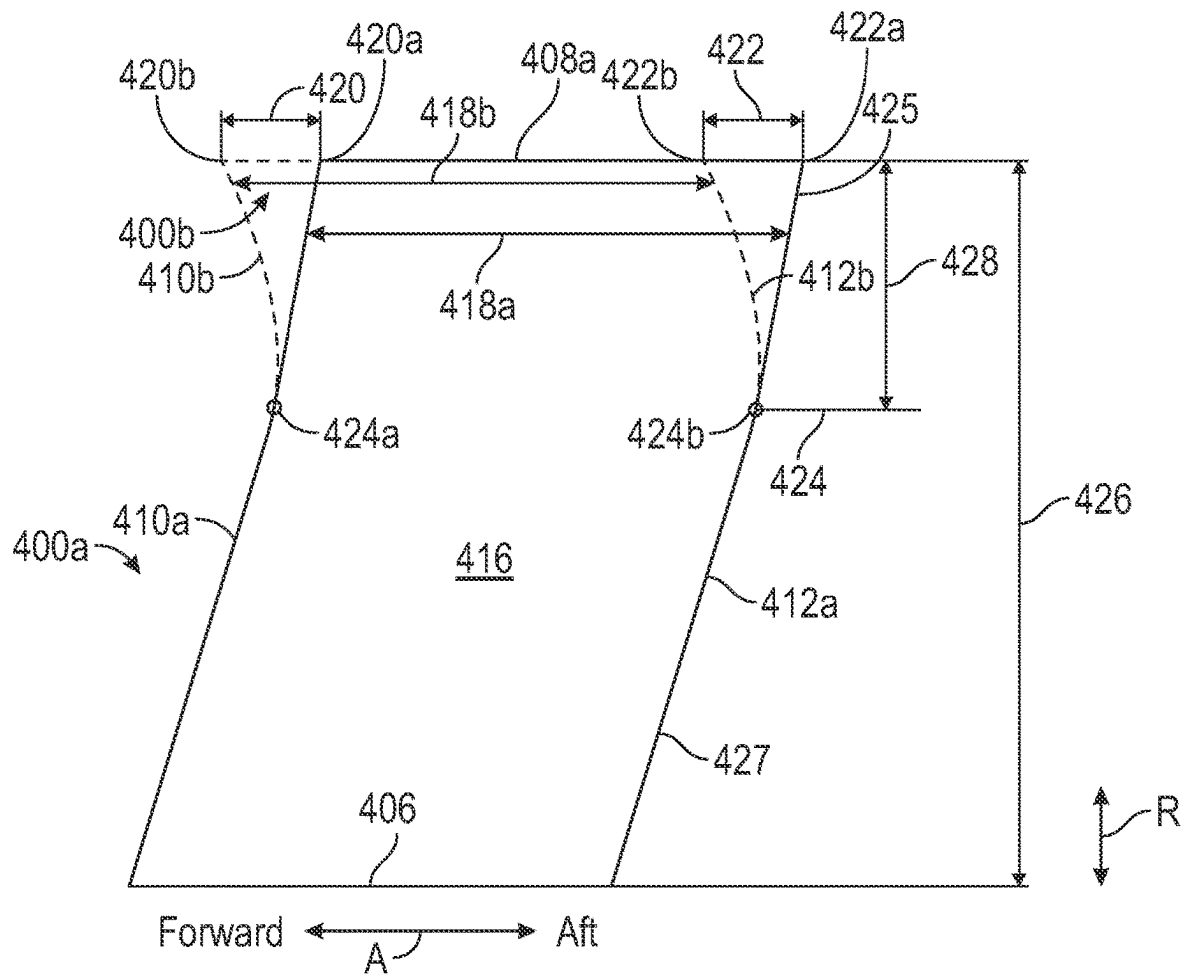


FIG. 4A

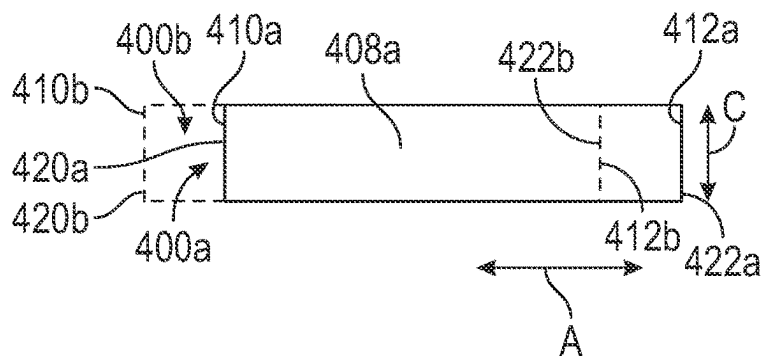


FIG. 4B

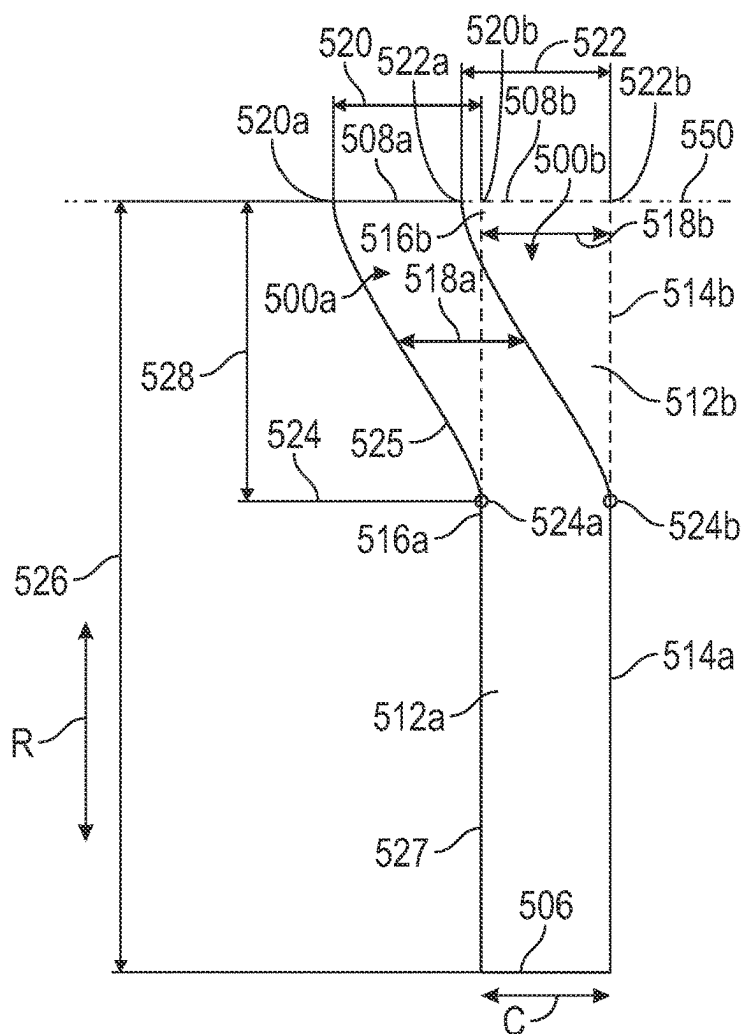


FIG. 5A

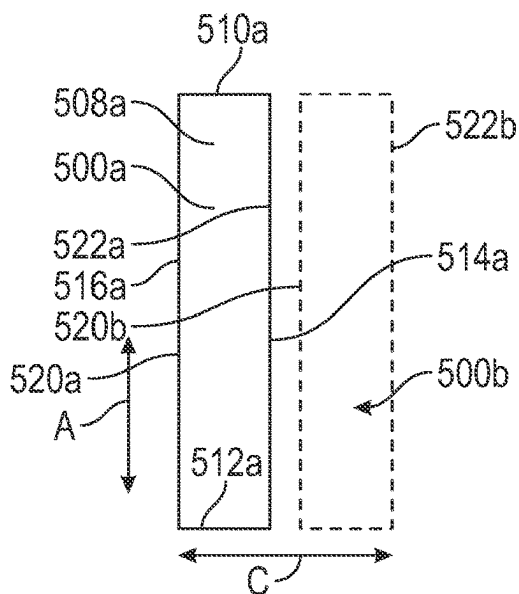


FIG. 5B

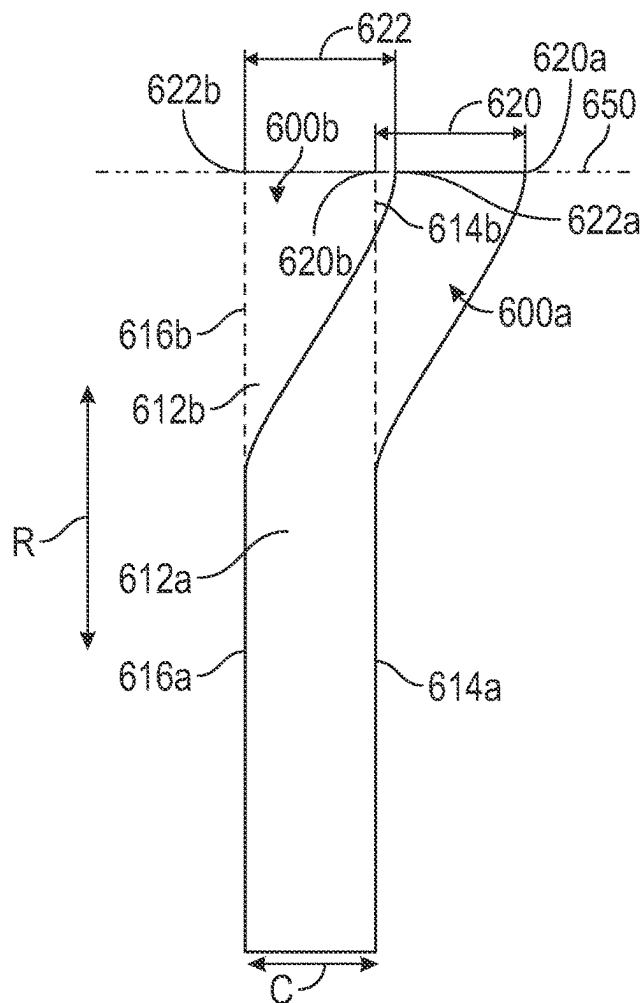


FIG. 6A

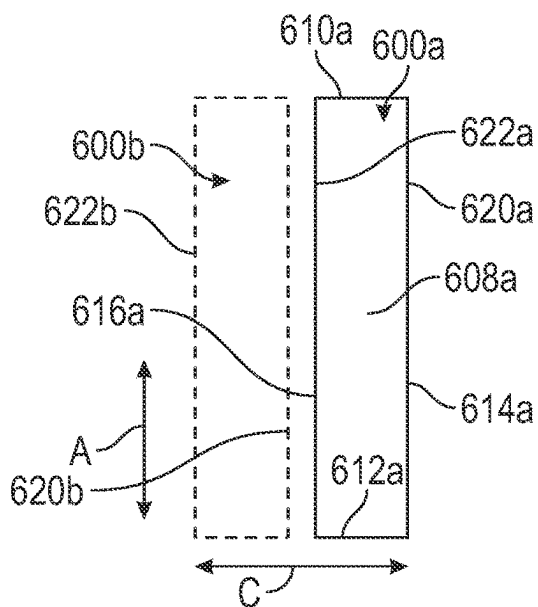


FIG. 6B

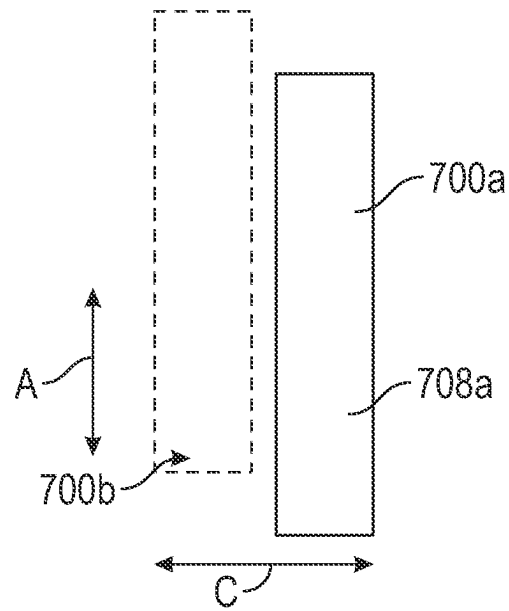


FIG. 7

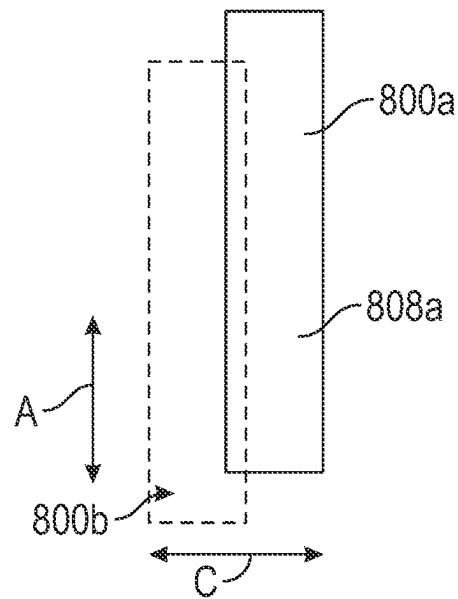


FIG. 8

1

ROTOR BLADE SYSTEM FOR TURBINE ENGINES

TECHNICAL FIELD

The present disclosure relates generally to a rotor blade system for turbine engines.

BACKGROUND

A turbine engine generally includes a fan and a core arranged in flow communication with one another. The core includes one or more turbines and one or more compressors. The turbines and compressors include one or more stages, with each stage including rotor blades and stator vanes.

BRIEF DESCRIPTION OF THE DRAWINGS

The foregoing and other features and advantages will be apparent from the following, more particular, description of various exemplary embodiments, as illustrated in the accompanying drawings, wherein like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

FIG. 1 illustrates a schematic, cross-sectional view of a ducted, indirect-drive gas turbine engine, taken along a longitudinal centerline axis of the engine, according to the present disclosure.

FIG. 2 illustrates a schematic, perspective view of a rotor blade system, according to the present disclosure.

FIG. 3A illustrates a schematic, suction side view of a rotor blade, which may be a rotor blade of the rotor blade system of FIG. 2, according to the present disclosure.

FIG. 3B illustrates a schematic, blade tip view of the rotor blade of FIG. 3A, according to the present disclosure.

FIG. 4A illustrates a schematic, suction side view of a rotor blade, which may be a rotor blade of the rotor blade system of FIG. 2, according to the present disclosure.

FIG. 4B illustrates a schematic, blade tip view of the rotor blade of FIG. 4A, according to the present disclosure.

FIG. 5A illustrates a schematic, trailing edge view of a rotor blade, which may be a rotor blade of the rotor blade system of FIG. 2, according to the present disclosure.

FIG. 5B illustrates a schematic, blade tip view of the rotor blade of FIG. 5A, according to the present disclosure.

FIG. 6A illustrates a schematic, trailing edge view of a rotor blade, which may be a rotor blade of the rotor blade system of FIG. 2, according to the present disclosure.

FIG. 6B illustrates a schematic, blade tip view of the rotor blade of FIG. 6A, according to the present disclosure.

FIG. 7 illustrates a schematic, blade tip view of a rotor blade having a combination of blade sweeps, which may be the rotor blade system of FIG. 2, according to the present disclosure.

FIG. 8 illustrates a schematic, blade tip view of a rotor blade having a combination of blade sweeps, which may be the rotor blade system of FIG. 2, according to the present disclosure.

DETAILED DESCRIPTION

Features, advantages, and embodiments of the present disclosure are set forth or apparent from a consideration of the following detailed description, drawings, and claims. Moreover, the following detailed description is exemplary and intended to provide further explanation without limiting the scope of the disclosure as claimed.

2

Various embodiments of the present disclosure are discussed in detail below. While specific embodiments are discussed, this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without departing from the spirit and the scope of the present disclosure.

As used herein, the terms “first” and “second” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

The terms “forward” and “aft” refer to relative positions within a turbine engine or a vehicle, and refer to the normal operational attitude of the turbine engine or the vehicle. For example, with regard to a turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or an exhaust.

As used herein, the terms “low,” “mid” (or “mid-level”), and “high,” or their respective comparative degrees (e.g., “lower” and “higher”, where applicable), when used with compressor, turbine, shaft, fan, or turbine engine components, each refers to relative pressures, relative speeds, relative temperatures, and/or relative power outputs within an engine unless otherwise specified. For example, a “low power” setting defines the engine configured to operate at a power output lower than a “high power” setting of the engine, and a “mid-level power” setting defines the engine configured to operate at a power output higher than a “low power” setting and lower than a “high power” setting. The terms “low,” “mid” (or “mid-level”), or “high” in such aforementioned terms may additionally, or alternatively, be understood as relative to minimum allowable speeds, pressures, or temperatures, or minimum or maximum allowable speeds, pressures, or temperatures relative to normal, desired, steady state, etc., operation of the engine.

The terms “coupled,” “fixed,” “attached,” “connected,” and the like, refer to both direct coupling, fixing, attaching, or connecting, as well as indirect coupling, fixing, attaching, or connecting through one or more intermediate components or features, unless otherwise specified herein. The terms include integral and unitary configurations (e.g., blisk rotor blade systems).

The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

As used herein, the terms “axial” and “axially” refer to directions and orientations that extend substantially parallel to a centerline of the turbine engine. Moreover, the terms “radial” and “radially” refer to directions and orientations that extend substantially perpendicular to the centerline of the turbine engine. In addition, as used herein, the terms “circumferential” and “circumferentially” refer to directions and orientations that extend arcuately about the centerline of the turbine engine.

Turbine engines include rotor blade systems that comprise a plurality of rotor blades in turbines and compressors. During rotation of the rotor blades, a vibratory mode exists where all of the rotor blades vibrate. By mistuning the rotor blades, a uniform vibratory mode of the rotor blades can be disrupted, thus, reducing the overall vibration level in the rotor blade system. The mistuning causes a non-uniformity in the vibratory mode, which reduces the vibratory response. The mistuning takes the form of a blade sweep in the axial direction, the circumferential direction, or both the axial and

the circumferential direction. The blade sweep occurs without addition or removal of material from the rotor blade or with minimal changes to the geometry of the airfoil of the rotor blade (e.g., within ten percent of the chord length or within twenty-five percent of the blade thickness, or both).

FIG. 1 shows a schematic, cross-sectional view of a turbine engine 100, taken along a longitudinal centerline axis 102 of the turbine engine 100, according to an embodiment of the present disclosure. The turbine engine 100 includes, in a downstream serial flow relationship, a fan section 104 including a fan 106, a compressor section 108 including a booster or a low-pressure (LP) compressor 110 and a high-pressure (HP) compressor 112, a combustion section 114 including a combustor 116, a turbine section 118 including a high-pressure (HP) turbine 120, a low-pressure (LP) turbine 122, and an exhaust nozzle 124. As shown in FIG. 1, the turbine engine 100 defines an axial direction A, a radial direction R, and a circumferential direction C. As shown and described herein, the turbine engine 100 is a ducted, indirect-drive, gas turbine engine 100.

The fan section 104 includes a fan casing 126, which is secured to a nacelle (omitted for clarity) surrounding the fan 106. The fan 106 includes a plurality of fan blades 128 disposed radially about the longitudinal centerline axis 102. The HP compressor 112, the combustor 116, and the HP turbine 120 form an engine core 130 of the turbine engine 100, which generates combustion gases. The engine core 130 is surrounded by a core casing 132, which is coupled to the fan casing 126. The fan casing 126 is supported relative to the turbomachine by circumferentially spaced outlet guide vanes 134.

A high-speed shaft 136, also referred to herein as a high-pressure shaft 136, is disposed coaxially about the longitudinal centerline axis 102 of the turbine engine 100 and drivingly connects the HP turbine 120 to the HP compressor 112. A low-speed shaft 138, also referred to herein as a low-pressure shaft 138, which is disposed coaxially about the longitudinal centerline axis 102 of the turbine engine 100 and within the larger diameter, annular, high-speed shaft 136, drivingly connects the LP turbine 122 to the LP compressor 110 and the fan 106 (either directly or indirectly through a gearbox assembly 140). The high-speed shaft 136 and the low-speed shaft 138 are rotatable about the longitudinal centerline axis 102.

The LP compressor 110 and the HP compressor 112, respectively, include a respective plurality of compressor stages 142, 144, in which a respective set of compressor blades 146, 148 rotate relative to a respective set of compressor vanes 150, 152 to compress or to pressurize gas entering through an inlet 154. Each compressor stage 144 of the HP compressor 112 includes multiple compressor blades 148 provided on a rotor disk 156 (or the blades and the disk are integrated together, referred to as a blisk), also referred to herein as rotor compressor blades 148 or rotor blades 148. Each compressor blade 148 extends radially outwardly relative to the longitudinal centerline axis 102, from a blade platform to a blade tip. Compressor vanes 152, also referred to herein as stator vanes 152, are positioned upstream/downstream of and adjacent to rotor compressor blades 148. The rotor disk 156 for a stage of compressor blades 148 is mounted to the high-speed shaft 136. The compressor stage 144 of the HP compressor 112 may refer to a single disk of rotor compressor blades 148 or may refer to both the single disk of rotor compressor blades 148 and an adjacent single disk of stator vanes 152. Either meaning can apply within the context of this disclosure without loss of clarity. The same description applies to each compressor stage 142 of the

LP compressor 110 (e.g., each compressor stage 142 of the LP compressor 110 includes multiple compressor blades 146, also referred to as rotor blades 146, and stator compressor vanes 150).

The HP turbine 120 has one or two turbine stages 158. In a single turbine stage 158, turbine blades 160 are provided on a rotor disk 162, also referred to herein as rotor blades 160. Each turbine blade 160 extends radially outwardly relative to the longitudinal centerline axis 102, from a blade platform to a blade tip. The HP turbine 120 can also include stator turbine vanes 164, also referred to as stator turbine nozzles. The HP turbine 120 may have an upstream nozzle adjacent an exit of the combustor 116 and a downstream nozzle aft of the rotor (e.g., turbine blades 160) or the HP turbine 120 may have a nozzle upstream of the rotor blades (e.g., turbine blades 160) or downstream of the rotor blades.

Air exiting the HP turbine 120 enters the LP turbine 122, which has a plurality of turbine stages 166 of rotor blades 168. The LP turbine 122 can have three, four, five, or six stages. In a single LP turbine stage 166 (containing a plurality of rotor blades 168 coupled to the low-speed shaft 138), the rotor blades 168 are provided on a rotor disk (connected to the low-speed shaft 138) and extend radially outwardly relative to the longitudinal centerline axis 102, from a blade platform to a blade tip. The LP turbine 122 can also include stator turbine vanes 170, also referred to as a stator turbine nozzle. The LP turbine 122 may have both an upstream nozzle and a downstream nozzle aft of a turbine stage 166, followed by the exhaust nozzle 124.

During operation of the turbine engine 100, a volume of air A1 enters the turbine engine 100 through an inlet 172 of the fan casing 126. As the volume of air A1 passes through the fan section 104 and across the fan blades 128, a first portion of air A2 of the air A1 is directed or routed into a bypass air flow passage 174 and a second portion of air A3 of the air A1 is directed or routed into the inlet 154 at an upstream section of a core air flow passage 176. The ratio between the first portion of air A2 and the second portion of air A3 is commonly known as a bypass ratio. The pressure of the second portion of air A3 is then increased as it is routed through the HP compressor 112 and into the combustion section 114, where the highly pressurized air is mixed with fuel and burned to provide combustion gases 178.

The combustion gases 178 are routed into the HP turbine 120 and expanded through the HP turbine 120 where a portion of thermal and/or kinetic energy from the combustion gases 178 is extracted via sequential stages of the HP turbine 120 turbine vanes 164 and rotor blades 160, which are coupled to the high-speed shaft 136, thus causing the high-speed shaft 136 to rotate, thereby supporting operation of the HP compressor 112. The combustion gases 178 are then routed into the LP turbine 122 and expanded through the LP turbine 122. Here, a second portion of thermal and kinetic energy is extracted from the combustion gases 178 via sequential stages of the LP turbine 122 turbine vanes 170 and the LP turbine rotor blades 168 that are coupled to the low-speed shaft 138, thus, causing the low-speed shaft 138 to rotate. The rotation of the low-speed shaft 138 thereby supports operation of the LP compressor 110 and rotation of the fan 106 (via the gearbox assembly 140, when present).

The combustion gases 178 are subsequently routed through the exhaust nozzle 124 to provide propulsive thrust. Simultaneously, the pressure of the first portion of air A2 is substantially increased as the first portion of air A2 is routed

5

through the bypass air flow passage **174** before being exhausted from a fan nozzle exhaust **180**, also providing propulsive thrust.

The turbine engine **100** is by way of example only. In other embodiments, the gas turbine engine may have any other suitable configuration, including, for example, any other suitable number or configurations of shafts or spools, fan blades, turbines, compressors, or a combination thereof. The gearbox assembly **140** may have any suitable configuration, including, for example, a star gear configuration, a planet gear configuration, a single-stage, a multi-stage, epicyclic, non-epicyclic, etc., as detailed further below. The gearbox assembly **140** may have a gear ratio in a range of 3:1 to 4:1, 3:5 to 4:1, 3.25:1 to 3.5:1, or 4:1 to 5:1. The fan assembly may be any suitable fixed-pitched assembly or variable-pitched assembly. The turbine engine **100** may include additional components not shown in FIG. 1, such as rotor blades, stator vanes, etc. The fan assembly may be configured in any other suitable manner (e.g., as a fixed pitch fan) and further may be supported using any other suitable fan frame configuration. Aspects of the present disclosure may be incorporated into any other suitable turbine engine, including, but not limited to, turbofan engines, propfan engines, turbojet engines, turboprop, and turboshaft engines, aviation-based turbine engines, marine-based turbine engines, land-based turbine engines, industrial turbine engines, power generation turbine engines, etc.

As noted, the high-pressure compressors and turbines and the low-pressure compressors and turbines include one or more stages each having two types of blades: stator blades and rotor blades. The blades, whether stator or rotor, are arranged in a circumferential manner about the longitudinal centerline axis **102** (FIG. 1). That is, each stage of the compressor and the turbine is associated with a ring or a circumferential row of stator blades and a ring or a circumferential row of rotor blades. The rotor blades or the stator blades are axisymmetric about the longitudinal centerline axis **102**.

FIG. 2 is a forward perspective view of a rotor blade system **200** isolated from the turbine engine **100** (FIG. 1), according to the present disclosure. The rotor blade system **200** defines a circumferential direction **C** and has a rotor **202** and a plurality of rotor blades **204** mounted thereon. The rotor blade system **200** rotates about a longitudinal centerline axis **12**, which is aligned with the longitudinal centerline axis **102** of the turbine engine **100** (FIG. 1). The rotor blade system **200** may be one or more of the fan **106** with fan blades **128**, the LP compressor **110** with rotor blades **146**, the HP compressor **112** with rotor blades **148**, the LP turbine **122** with rotor blades **168**, or the HP turbine **120** with rotor blades **160**. The rotor blade system **200** is equally applicable to industrial gas turbines (IGTs) or power generation turbines. The present rotor blade system is not limited only to turbine engines but can be used in any airfoil system.

The plurality of rotor blades **204** are uniformly and circumferentially spaced about the rotor **202**. The rotor blades **204** may be integrally formed with the rotor **202** or may be connected thereto. Each rotor blade **204** is formed of an airfoil and extends from a blade root **206** at the rotor **202** to a blade tip **208** and includes a leading edge **210** and a trailing edge **212**. As illustrated in FIG. 2, the airfoil of each rotor blade **204** may have a twist or contour. For example, each rotor blade **204** may be contoured or twisted with respect to a longitudinal blade axis **218** extending from the blade root **206** to the blade tip **208**. Each rotor blade **204** includes a pressure side **214** joined to a suction side **216** at

6

the leading edge **210** and at the trailing edge **212**. The pressure side **214** is concave and the suction side **216** is convex.

Adjacent rotor blades **204** have a passage length therebetween, also referred to as an inter-blade spacing therebetween. A pitch **P** is the inter-blade spacing and is defined as the distance between adjacent rotor blades **204** at the respective blade tips **208**. Adjacent rotor blades are defined as two rotor blades **204** with no intervening rotor blade **204** therebetween.

One or more of the rotor blades **204** may be mistuned without removal of material or addition of material from the rotor blades **204**, or with minimal removal or addition (e.g., within ten percent). That is, one or more of the rotor blades **204** may include a blade sweep extending from a blade tip toward a blade root. The blade sweep, also referred to as blade shift, achieves blade mistuning with no or minimal material removal or addition and no or minimal (e.g., within ten percent) change in blade weight.

In some examples, all rotor blades **204** may be mistuned according to one or more of the embodiments described with respect to FIGS. 3A to 8. In some examples, the mistuning of the rotor blades **204** may be a predetermined patterned arrangement, which may be a predetermined grouped arrangement or a predetermined alternating arrangement. That is, for example, when fewer than all rotor blades **204** are mistuned, the mistuning may be according to a predetermined arrangement. For example, the rotor blades **204** may take on any number of patterns including, but not limited to, “AB” patterns, “AAB” patterns, “ABB” patterns, “ABC” patterns, “AABC” patterns, “AABBC” patterns, “AABBCC” patterns, etc., or combinations thereof, where A is a base blade, and each of B and C are mistuned blades. Although the pattern describes only two types of mistuned blades, more types may be provided (e.g., D, E, F, etc.) such that multiple exemplary embodiments of mistuned blades may be present in the rotor blade system **200**. B and C (and D, E, F, etc.) represent differently mistuned rotor blades (e.g., different rotor blades according to FIGS. 3A to 8). In another example, the predetermined grouped arrangement may comprise a subset of mistuned rotor blades arranged adjacent to one another and located at a single location about the rotor **202** of the rotor blade system **200**. For example, a predetermined group of mistuned rotor blades may be located from, with respect to a clockwise direction of a clock arrangement, a twelve o’clock position to three o’clock position with the base rotor blades located from the three o’clock position (clockwise) to the twelve o’clock position. This is simply exemplary, and any grouping of mistuned rotor blades is contemplated.

FIGS. 3A to 8 illustrate exemplary embodiments of rotor blades that may be employed as one or more of the rotor blades **204**. As described below, the blades include blade sweeps in one or more directions. In some examples, the blades include axial blade sweeps, circumferential blade sweeps, or both axial blade sweeps and circumferential blade sweeps. The mistuning provided in FIGS. 3A to 8 allows for aero-damping to reduce non-synchronous responses. The rotor blades of FIGS. 3A to 8 are illustrated schematically and, thus, do not illustrate the curvature as shown with respect to FIG. 2. However, this is for simplicity in description only and it is understood that the rotor blades of FIGS. 3A to 8 also include curvature such as shown in FIG. 2.

FIGS. 3A, 3B, 4A, and 4B illustrate rotor blades exhibiting axial blade sweep of a leading edge and a trailing edge. As described in more detail below, the axial blade sweep is

defined as an axial change of the leading edge or trailing edge with respect to a leading edge or a trailing edge of a base blade, respectively. The axial blade sweep results in a leading edge or a trailing edge of the rotor blade being located farther forward or farther aft as compared to the base blade. In the embodiments described herein, the axial sweep occurs over a portion of a span length of the rotor blade. That is, the base blade is substantially linear with respect to a lower portion of the leading edge or trailing edge and the swept blade is nonlinear with respect to the leading edge or trailing edge above the lower portion.

FIGS. 3A and 3B illustrate a rotor blade 300a having a blade tip 308a, a blade root 306, a leading edge 310a, and a trailing edge 312a. A chord length 318a is defined between the leading edge 310a and the trailing edge 312a at the blade tip 308a. As shown in FIG. 2, the chord length 318a may not be a linear measurement due to the curvature of the rotor blade 300a. The view of FIG. 3A illustrates a suction side 316 of the rotor blade 300a. FIG. 3B illustrates a view of the blade tip 308a. As illustrated in FIGS. 3A and 3B, the blade tip 308a of the leading edge 310a is swept in the axial direction A (as defined in FIGS. 1 and 2) toward the forward end of the turbine engine 100 (FIG. 1). The axial sweep is in relation to a base blade 300b. The base blade 300b has a base blade leading edge 310b, a base blade trailing edge 312b, a base blade tip 308b, and a base blade root. The base blade tip 308b and the blade tip 308a are coplanar. The base blade root and the blade root 306 are coplanar. At the blade root 306, the leading edge and the trailing edge of each of the base blade 300b and the rotor blade 300a are at the same axial location and there is no axial sweep.

FIG. 3A illustrates the leading edge 310a substantially collinear with respect to the base blade leading edge 310b along a lower portion 327 of the leading edge 310a that extends from the blade root 306 to a sweep point 324a. After the sweep point 324a, along an upper portion 325 of the leading edge 310a, the leading edge 310a is not collinear with respect to the base blade leading edge 310b. Thus, the sweep point 324a defines a point at which the leading edge 310a diverges from the base blade leading edge 310b. Likewise, the trailing edge 312a is substantially collinear with respect to the base blade trailing edge 312b along a lower portion of the trailing edge 312a that extends from the blade root 306 to a sweep point 324b. After the sweep point 324b and extending from the sweep point 324b to the blade tip 308a, the trailing edge 312a is not collinear with respect to the base blade trailing edge 312b. Thus, the sweep point 324b defines a point at which the trailing edge 312a diverges from the base blade trailing edge 312b.

As mentioned, the leading edge 310a is swept in the axial direction A toward the forward end of the turbine engine 100 (FIG. 1). "Axial sweep" means that the leading edge 310a is axially farther in the forward direction than the base blade leading edge 310b. As noted above, the axial sweep of the leading edge 310a occurs from the sweep point 324a to the blade tip 308a. Likewise, the trailing edge 312a is swept in the axial direction A toward the forward end of the turbine engine 100 (FIG. 1) as compared to the base blade trailing edge 312b. The axial sweep of the trailing edge 312a occurs from the sweep point 324b to the blade tip 308a. The sweep of the leading edge 310a and the trailing edge 312a, is such that the chord length 318a defined between the leading edge 310a and the trailing edge 312a is maintained constant with respect to a base blade chord length 318b of the base blade 300b. A constant chord length 318a is a chord length that is within ten percent (either greater or lesser) of the base blade chord length 318b defined between the base blade leading

edge 310b and the base blade trailing edge 312b. In some examples, the chord length 318a is equal to the base blade chord length 318b. By maintaining a constant chord length 318a, the rotor blade 300a performs about the same amount (e.g., within ten percent) of work as the base blade 300b, while benefiting from the mistuning properties that disrupt the vibrational mode.

At the blade tip 308a, the amount that the leading edge 310a is swept in the axial direction A is equal to a leading edge axial chord sweep 320. The leading edge axial chord sweep 320 is defined between a leading tip edge 320a and a base blade leading tip edge 320b. Referring to FIGS. 3A and 3B, the leading tip edge 320a is a substantially circumferentially extending edge of the blade tip 308a at the leading edge 310a and the base blade leading tip edge 320b is a circumferentially extending edge of the base blade tip 308b at the base blade leading edge 310b. By "substantially circumferentially" it is meant that the edge extends more in the circumferential direction than in either of the radial or axial directions. Similarly, at the blade tip 308a, the amount that the trailing edge 312a is swept in the axial direction A is equal to a trailing edge axial chord sweep 322. The trailing edge axial chord sweep 322 is defined between a trailing tip edge 322a and a base blade trailing tip edge 322b. Referring to FIGS. 3A and 3B, the trailing tip edge 322a is a substantially circumferentially extending edge of the blade tip 308a at the trailing edge 312a and the base blade trailing tip edge 322b is a circumferentially extending edge of the base blade tip 308b at the base blade trailing edge 312b.

The leading edge axial chord sweep 320 and the trailing edge axial chord sweep 322 may increase from the sweep points 324a, 324b to the blade tip 308a. That is, the distance between the leading edge 310a and the base blade leading edge 310b may increase from the sweep point 324a to the blade tip 308a or the distance may be constant. Likewise with the trailing edge axial chord sweep 322. In some examples, the leading edge axial chord sweep 320 and the trailing edge axial chord sweep 322 are equal. In some examples, the leading edge axial chord sweep 320 and the trailing edge axial chord sweep 322 are different, while maintaining the chord length 318a within ten percent of the base blade chord length 318b. The leading edge axial chord sweep 320 is from greater than zero percent to fifteen percent of the chord length 318a, and, preferably, from five percent to ten percent of the chord length 318a. The trailing edge axial chord sweep 322 is from greater than zero percent to fifteen percent of the chord length 318a, preferably, from five percent to ten percent of the chord length 318a.

A span length 326 is defined between the blade tip 308a and the blade root 306. The axial sweep of the leading edge 310a and the trailing edge 312a may extend from an axis 324 to the blade tip 308a. The axis 324 extends through the sweep point 324a (e.g., the point at which the leading edge 310a and the trailing edge 312a are no longer swept and are collinear with, respectively, the base blade leading edge 310b and the base blade trailing edge 312b). The percentage of the span length 326 of the rotor blade 300a that exhibits the axial blade sweep is defined by a radial distance 328 between the blade tip 308a and the axis 324. The radial distance 328 is from greater than zero percent to forty percent of the span length 326. In some examples, the radial distance is from five percent to twenty-five percent of the span length 326 of the rotor blade 300a. In some examples, the radial distance is twenty-five percent of the span length 326 of the rotor blade 300a.

FIGS. 4A and 4B illustrate a rotor blade 400a, having a blade tip 408a, a blade root 406, a leading edge 410a, and

a trailing edge **412a**. FIG. 4A illustrates a suction side **416** of the rotor blade **400a**. FIG. 4B illustrates a view of the blade tip **408a**. The rotor blade **400a** is equal to the rotor blade **300a**, except that the leading edge **410a** and the trailing edge **412a** of the rotor blade **400a** are swept in the axial direction A toward the aft end of the turbine engine **100** (FIG. 1). Accordingly, the full description of FIGS. 3A and 3B applies to FIGS. 4A and 4B, except for the direction of the axial sweep.

That is, the leading edge **410a** and the trailing edge **412a** are swept in the axial direction by a leading edge axial chord sweep **420** and a trailing edge axial chord sweep **422**, respectively, as compared to a base blade leading edge **410b** and a base blade trailing edge **412b** of a base blade **400b**, respectively. The axial chord sweep is defined according to the definition of FIG. 3A. For example, the leading edge axial chord sweep **420** is defined between a leading tip edge **420a** and a base blade leading tip edge **420b**. Referring to FIGS. 4A and 4B, the leading tip edge **420a** is a circumferentially extending edge of the blade tip **408a** at the leading edge **410a** and the base blade leading tip edge **420b** is a circumferentially extending edge of the base blade tip at the base blade leading edge **410b**. The trailing edge axial chord sweep **422** is defined between a trailing tip edge **422a** and a base blade trailing tip edge **422b**. Referring to FIGS. 4A and 4B, the trailing tip edge **422a** is a circumferentially extending edge of the blade tip **408a** at the trailing edge **412a** and the base blade trailing tip edge **422b** is a circumferentially extending edge of the base blade tip at the base blade trailing edge **412b**.

In some examples, the leading edge axial chord sweep **420** and the trailing edge axial chord sweep **422** may be equal. In some examples, the leading edge axial chord sweep **420** and the trailing edge axial chord sweep **422** are different, while maintaining a constant chord length, e.g., maintaining a chord length **418a** within ten percent of a base chord length **418b**. As noted previously, by maintaining a constant chord length **418a**, the rotor blade **400a** performs about the same amount (e.g., within ten percent) of work as the base blade **400b**, while benefiting from the mistuning properties that disrupt the vibrational mode. The leading edge axial chord sweep **420** is from greater than zero percent to fifteen percent of the chord length **418a**, preferably, from five percent to ten percent of the chord length **418a**. The trailing edge axial chord sweep **422** is from greater than zero percent to fifteen percent of the chord length **418a**, preferably from five percent to ten percent of the chord length **418a**.

As in FIG. 3A, a sweep point **424b** exists along the trailing edge **412a** of the rotor blade **400a**. The trailing edge **412a** is substantially collinear with respect to the base blade trailing edge **412b** along a lower portion **427** of the trailing edge **412a** that extends from the blade root **406** to a sweep point **424b**. After the sweep point **424b** along an upper portion **425**, the trailing edge **412a** is not collinear with respect to the base blade trailing edge **412b**. Thus, the sweep point **424b** defines a point at which the trailing edge **412a** diverges from the base blade trailing edge **412b**. A sweep point **424a** is located along the leading edge **410a** and is defined in the same manner described previously with respect to both FIGS. 3A and 4A.

A span length **426** is defined between the blade tip **408a** and the blade root **406**. The axial sweep of the leading edge **410a** and the trailing edge **412a** may extend from an axis **424** to the blade tip **408a**. The axis **424** extends through the sweep point **424b** (e.g., the point at which the leading edge **410a** and the trailing edge **412a** are no longer swept and are

collinear with, respectively, the base blade leading edge **410b** and the base blade trailing edge **412b**). The percentage of the span length **426** of the rotor blade **400a** that exhibits the axial blade sweep is defined by a radial distance **428**. The radial distance **428** is defined between the blade tip **408a** and an axis **424** that extends through the point at which the leading edge **410a** and the trailing edge **412a** are no longer swept and are collinear with, respectively, the base blade leading edge **410b** and the base blade trailing edge **412b**. The radial distance may be from greater than zero percent to forty percent of the span length **426** of the rotor blade **400a**. In some examples, the radial distance is from five percent to twenty-five percent of the span length of the rotor blade **400a**. In some examples, the radial distance is twenty-five percent of the span length of the rotor blade **400a**.

FIGS. 5A, 5B, 6A, and 6B illustrate rotor blades exhibiting tangential sweep of a suction side and a pressure side in the circumferential direction. That is, the rotor blades of FIGS. 5A, 5B, 6A, and 6B exhibit a sweep in a direction of the pitch P (FIG. 2) and therefore extend into the inter-blade spacing.

FIG. 5A illustrates a rotor blade **500a** having a blade tip **508a**, a blade root **506**, a leading edge **510a**, a trailing edge **512a**, a pressure side **514a**, and a suction side **516a**. The view of FIG. 5A illustrates an aft end view of the rotor blade **500a** such that the leading edge is not visible in this view. FIG. 5B illustrates a view of the blade tip **508a** taken at a plane **550** such that only the blade tip **508a** and a base blade tip **508b** are visible in FIG. 5B. In the example of FIGS. 5A and 5B, the suction side **516a** is swept in the circumferential direction C (as defined in FIGS. 1 and 2) in the direction opposite of rotation of the turbine engine **100** (FIG. 1). The circumferential sweep is in relation to a base blade **500b**. The base blade **500b** has a base blade suction side **516b**, a base blade pressure side **514b**, a base blade trailing edge **512b**, the base blade tip **508b**, and a base blade root. As in the aspects of FIGS. 3 and 4, the base blade tip **508b** and the base blade root are collinear with the blade tip **508a** and the blade root **506**, respectively. At the blade root, the leading edge and the trailing edge of each of the base blade **500b** and the rotor blade **500a** are at the same axial location and there is no axial sweep.

As mentioned, the suction side **516a** is swept in the circumferential direction C in the direction opposite of rotation of the turbine engine **100** (FIG. 1). "Circumferential sweep," means that the suction side **516a** is circumferentially farther in the direction opposite of rotation of the turbine engine **100** (FIG. 1) as compared to the base blade suction side **516b**. Likewise, the pressure side **514a** is swept in the circumferential direction C in the direction opposite of the rotation of the turbine engine **100** (FIG. 1) as compared to the base blade pressure side **514b**. The sweep of the suction side **516a** and the pressure side **514a** is such that a chord length (as defined with respect to FIGS. 3A and 3B) is constant (e.g., within ten percent of) between the rotor blade **500a** and the base blade **500b** (e.g., as described with respect to FIGS. 3A to 4B). In some examples, a blade thickness **518a** defined between the suction side **516a** and the pressure side **514a** is within twenty-five percent of a blade thickness **518b** defined between the base blade suction side **516b** and the base blade pressure side **514b**. In some examples, the blade thickness **518a** is equal to the blade thickness **518b**.

FIG. 5A illustrates the suction side **516a** substantially collinear with respect to the base blade suction side **516b** along a lower portion **527** of the suction side **516a** that extends from the blade root **506** to a sweep point **524a**. After

11

the sweep point **524a**, along an upper portion **525** of the suction side **516a**, the suction side **516a** is not collinear with respect to the base blade suction side **516b**. Thus, the sweep point **524a** defines a point at which the suction side **516a** diverges from the base blade suction side **516b**. Likewise, the pressure side **514a** is substantially collinear with respect to the base blade pressure side **514b** along a lower portion of the pressure side **514a** that extends from the blade root **506** to a sweep point **524b**. After the sweep point **524b** and extending from the sweep point **524b** to the blade tip **508a**, the pressure side **514a** is not collinear with respect to the base blade pressure side **514b**. Thus, the sweep point **524b** defines a point at which the pressure side **514a** diverges from the base blade pressure side **514b**.

The circumferential sweep of the suction side **516a** and the pressure side **514a** may extend from an axis **524** to the blade tip **508a**. The axis **524** extends through the point at which the suction side **516a** and the pressure side **514a** are no longer swept and are collinear with the base blade suction side **516b** and the base blade pressure side **514b**, respectively. At the blade tip **508a**, the amount that the suction side **516a** is swept in the circumferential direction **C** is equal to a suction side circumferential sweep **520**. The suction side circumferential sweep **520** is defined between a suction side tip edge **520a** and a base blade suction side tip edge **520b**. Referring to FIGS. **5A** and **5B**, the suction side tip edge **520a** is a substantially axially extending edge of the blade tip **508a** at the suction side **516a** and the base blade suction side tip edge **520b** is a substantially axially extending edge of the base blade tip **508b** at the base blade suction side **516b**. By “substantially axially” it is meant that the edge extends more in the axially direction than in either of the radial or circumferential directions. Similarly, at the blade tip **508a**, the amount that the pressure side **514a** is swept in the circumferential direction **C** is equal to a pressure side circumferential sweep **522**. The pressure side circumferential sweep **522** is defined between a pressure side tip edge **522a** and a base blade pressure side tip edge **522b**. Referring to FIGS. **5A** and **5B**, the pressure side tip edge **522a** is a substantially axially extending edge of the blade tip **508a** at the pressure side **514a** and the base blade pressure side tip edge **522b** is a substantially axially extending edge of the base blade tip **508b** at the base blade pressure side **514b**.

The suction side circumferential sweep **520** and the pressure side circumferential sweep **522** may be equal. The suction side circumferential sweep **520** is from greater than zero percent to twenty-five percent of the pitch (e.g., FIG. **2**, pitch **P**). The pressure side circumferential sweep **522** is from greater than zero percent to twenty-five percent of the pitch. In some examples, the suction side circumferential sweep **520** or the pressure side circumferential sweep **522**, or both, is from two percent to ten percent. As in FIG. **2**, the pitch, is the inter-blade spacing defined as the distance between adjacent rotor blades **500a** at the respective blade tips. Thus, the sweep (either as described with respect to FIG. **5A** or **6A**) is a percentage of the inter-blade spacing between the rotor blade **500a** and an adjacent rotor blade upstream or downstream (depending on the direction of sweep). Adjacent means two rotor blades with no intervening rotor blade therebetween.

A span length **526** is defined between the blade tip **508a** and the blade root **506**. The percentage of the span length **526** of the rotor blade **500a** that exhibits the axial blade sweep is defined by a radial distance **528**. The radial distance **528** is defined between the blade tip **508a** and the axis **524**. The radial distance **528** is from greater than zero percent to forty percent of the span length **526**. In some examples, the

12

radial distance is from five percent to twenty-five percent of the span length of the rotor blade **500a**. In some examples, the radial distance is twenty-five percent of the span length of the rotor blade **500a**.

FIGS. **6A** and **6B** illustrate a rotor blade **600a**. The rotor blade **600a** has a blade tip **608a**, a leading edge **610a**, a trailing edge **612a**, a pressure side **614a**, and a suction side **616a**. FIG. **6A** illustrates an aft end view of the rotor blade **600a** and a base blade **600b** such that the trailing edge **612a** of the rotor blade **600a** and a trailing edge **612b** of the base blade are visible and the leading edge of each blade is not visible in this view. FIG. **6B** illustrates a view of the blade tip **608a** taken at a plane **650** such that only the blade tip **608a** and a base blade tip are visible in FIG. **6B**. The rotor blade **600a** is the same as the rotor blade **500a** of FIGS. **5A** and **5B**, except that the suction side **616a** and the pressure side **614a** of the rotor blade **600a** are swept in the circumferential direction **C** in the direction of rotation of the turbine engine **100** (FIG. **1**). Accordingly, the full description of FIG. **5A** applies to FIG. **6A**, except for the direction of the circumferential sweep.

That is, the suction side **616a** and the pressure side **614a** are swept in the circumferential direction by a suction side circumferential sweep **620** and a pressure side circumferential sweep **622**, respectively, as compared to a base blade suction side **616b** and a base blade pressure side **614b** of a base blade **600b**, respectively. The circumferential sweep is defined according to the definition of FIG. **5A**. For example, the suction side circumferential sweep **620** is defined between a suction side tip edge **620a** and a base blade suction side tip edge **620b**. Referring to FIGS. **6A** and **6B**, the suction side tip edge **620a** is a substantially axially extending edge of the blade tip **608a** at the suction side **616a** and the base blade suction side tip edge **620b** is a substantially axially extending edge of the base blade tip at the base blade suction side **616b**. The pressure side circumferential sweep **622** is defined between a pressure side tip edge **622a** and a base blade pressure side tip edge **622b**. Referring to FIGS. **6A** and **6B**, the pressure side tip edge **622a** is a substantially axially extending edge of the blade tip **608a** at the pressure side **614a** and the base blade pressure side tip edge **622b** is a substantially axially extending edge of the base blade tip at the base blade pressure side **614b**.

The suction side circumferential sweep **620** and the pressure side circumferential sweep **622** may be equal. The suction side circumferential sweep **620** is from greater than zero percent to twenty-five percent. The pressure side circumferential sweep **622** is from greater than zero percent to twenty-five percent. In some examples, the suction side circumferential sweep **620** or the pressure side circumferential sweep **622**, or both, is from two percent to ten percent. The percentage of the rotor blade **600a** that exhibits the circumferential sweep is defined by a radial distance and a sweep point as defined in FIG. **5A**. The radial distance is from greater than zero percent to forty percent of the span length of the rotor blade **600a**. In some examples, the radial distance is from five percent to twenty-five percent of the span length of the rotor blade **500a**. In some examples, the radial distance is twenty-five percent of the span length of the rotor blade **500a**. As in prior examples, a chord length (as defined with respect to FIGS. **3A** and **3B**) is constant (e.g., within ten percent of) between the rotor blade **600a** and the base blade **600b** (e.g., as described with respect to FIGS. **3A** to **4B**). In some examples, a blade thickness (as defined with respect to FIGS. **5A** and **5B**) of the rotor blade **600a** is within twenty-five percent of a blade thickness of the base blade **600b** or is the same as the thickness of the base blade **600b**.

13

As illustrated in FIGS. 5B and 6B, the circumferential sweep is such that the blade tip of the rotor blades has no overlap with the base blade tip of the base blades. However, an overlap may be present, such as shown and described with respect to FIG. 8.

As described previously, FIGS. 3A to 4B illustrate an axial chord sweep of the rotor blades and FIGS. 5A to 6B illustrate a circumferential sweep of the rotor blades. Although the examples are described separately, any combination of the sweeps may be provided in one or more of the rotor blades (e.g., rotor blades 204 described with respect to FIG. 2).

For example, as shown in the top views of FIGS. 7 and 8 (which are views of a blade tip 708a and a blade tip 808a, respectively), a rotor blade may be swept both axially and circumferentially as compared to a base blade. Each of FIGS. 7 and 8 is taken at a plane extending through the blade tips such that only the blade tip 708a, 808a and a base blade tip are visible. The particular example of FIG. 7 illustrates an axially forward sweep and a circumferential sweep the direction of rotation of the engine of a rotor blade 700a as compared to a base blade 700b. The particular example of FIG. 8 illustrates an axially aft sweep and a circumferential sweep in the direction of rotation of the engine of a rotor blade 800a as compared to a base blade 800b. Furthermore, in the example of FIG. 7, the circumferential sweep is such that there is no overlap between the base blade 700b and the rotor blade 700a at the blade tip 708a, while, in the example of FIG. 8, the circumferential sweep is such that there is overlap between the base blade 800b and the rotor blade 800a at the blade tip 808a. Likewise, rotor blades may exhibit circumferential sweep in the opposite direction of rotation with either an axially forward sweep or an axially aft sweep. In this example, there may or may not be overlap between the base blade and the rotor blade at the blade tip.

These are just two exemplary combinations, and the combination may exhibit in other manners. For example, the rotor blades may have an axially aft sweep and a circumferential sweep in the direction of rotation of the engine, and an axially forward sweep and a circumferential sweep in the direction opposite of rotation of the engine, or an axially forward sweep and a circumferential sweep in the direction of rotation of the engine. In any of these combinations, the circumferential sweep may or may not include an overlap at the blade tip.

Furthermore, as noted previously, any of the examples of FIGS. 3A to 8 may be applied to one or more of the rotor blades 204 (FIG. 2). In some examples, the same swept blade arrangement is provided to all of the one or more of the rotor blades 204. In some examples, more than one, different, swept blade arrangement is provided to one or more of the rotor blades 204.

Although the base rotor blades herein are illustrated in FIGS. 3A to 8 as parallelepipeds, these images are schematic only and other shapes are contemplated. That is, for example, the base rotor blade may have a twist or other contour to the blade, such as shown and described with respect to FIG. 2. The rotor blade of the present disclosure is then swept from the base blade, maintaining any original twist or contour of the base blade.

Accordingly, the rotor blades of the present disclosure change the airfoil shape without meaningfully adding material or removing material. That is, the sweep described herein does not add material or remove material from the rotor blade. No meaningful addition of material or removal material means that the chord length of the swept rotor blade is within ten percent of the base blade, the blade thickness

14

of the swept rotor blade is within twenty-five percent of the base blade, or both the chord length of the swept rotor blade is within ten percent of the base blade and the blade thickness of the swept rotor blade is within twenty-five percent of the base blade.

As illustrated and described herein, the leading edge and the trailing edge are swept in the same direction (e.g., either forward or aft) and the pressure side and the suction side are swept in the same direction (e.g., either in the direction of rotation or opposite the direction of rotation). In some examples, however, the sweep may be in different directions as long as the chord (for axial sweep) or thickness (for circumferential sweep) is maintained within ten percent of that of the base blade.

Further aspects are provided by the subject matter of the following clauses.

A rotor blade system for a turbine engine having a longitudinal centerline axis includes a rotor configured to rotate about the longitudinal centerline axis, and a plurality of rotor blades coupled circumferentially around the rotor, each rotor blade having a leading edge, a trailing edge, a suction side, and a pressure side. At least one rotor blade of the plurality of rotor blades has a leading edge axial sweep of the leading edge as compared to a leading edge of a base blade of the plurality of rotor blades and a trailing edge axial sweep of the trailing edge as compared to a trailing edge of the base blade of the plurality of rotor blades.

The rotor blade system of the preceding clause, wherein the leading edge axial sweep is equal to the trailing edge axial sweep.

The rotor blade system of any preceding clause, wherein the leading edge axial sweep and the trailing edge axial sweep extend in a forward direction of the turbine engine.

The rotor blade system of any preceding clause, wherein the leading edge axial sweep and the trailing edge axial sweep extend in an aft direction of the turbine engine.

The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is an overlap between the blade tip and the base blade tip.

The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is no overlap between the blade tip and the base blade tip.

The rotor blade system of any preceding clause, wherein the base blade comprises a twist, and the at least one rotor blade comprises the same twist as that of the base blade.

The rotor blade system of any preceding clause, wherein the at least one rotor blade has no addition or removal of material or minimal addition or removal of material as compared to the base blade.

The rotor blade system of any preceding clause, wherein the at least one rotor blade comprises all of the plurality of rotor blades.

The rotor blade system of any preceding clause, wherein the at least one rotor blade comprises multiple rotor blades and the multiple rotor blades are arranged in a predetermined pattern about the rotor with a plurality of base blades.

The rotor blade system of any preceding clause, wherein the predetermined pattern is an alternating pattern.

The rotor blade system of any preceding clause, wherein the at least one rotor blade of the plurality of rotor blades further comprises a circumferential sweep in an inter-blade

15

spacing defined between the at least one rotor blade of the plurality of rotor blades and an adjacent rotor blade of the plurality of rotor blades.

The rotor blade system of any preceding clause, wherein a pitch is defined between a blade tip of adjacent rotor blades of the plurality of rotor blades, and the circumferential sweep is from greater than zero percent to twenty-five percent of the pitch.

The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a chord length between the leading edge and the trailing edge, and wherein the leading edge axial sweep or the trailing edge axial sweep is from greater than zero percent to fifteen percent of the chord length.

The rotor blade system of any preceding clause, wherein the leading edge axial sweep or the trailing edge axial sweep is from five percent to ten percent of the chord length.

The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a span length between a blade tip and a blade root, and the leading edge axial sweep or the trailing edge axial sweep is from greater than zero percent to forty percent of the span length.

The rotor blade system of any preceding clause, wherein the leading edge axial sweep or the trailing edge axial sweep is from five percent to twenty-five percent of the span length.

The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a chord length between the leading edge and the trailing edge, the base blade defines a base blade chord length between a leading edge of the base blade and a trailing edge of the base blade, and the chord length is within ten percent of the base blade chord length.

The rotor blade system of any preceding clause, wherein the chord length is equal to the base blade chord length.

The rotor blade system of any preceding clause, further comprising a circumferential sweep, wherein the circumferential sweep is in a direction of rotation of the turbine engine.

The rotor blade system of any preceding clause, further comprising a circumferential sweep, wherein the circumferential sweep is in a direction opposite of rotation of the turbine engine.

The rotor blade system of any preceding clause, further comprising a circumferential sweep, and wherein the leading edge axial sweep and the trailing edge axial sweep are in a forward direction of the turbine engine and the circumferential sweep is in a direction of rotation of the turbine engine.

The rotor blade system of any preceding clause, further comprising a circumferential sweep, and wherein the leading edge axial sweep and the trailing edge axial sweep are in a forward direction of the turbine engine and the circumferential sweep is in a direction opposite of rotation of the turbine engine.

The rotor blade system of any preceding clause, further comprising a circumferential sweep, and wherein the leading edge axial sweep and the trailing edge axial sweep are in an aft direction of the turbine engine and the circumferential sweep is in a direction of rotation of the turbine engine.

The rotor blade system of any preceding clause, further comprising a circumferential sweep, and wherein the leading edge axial sweep and the trailing edge axial sweep are in an aft direction of the turbine engine and the circumferential sweep is in a direction opposite of rotation of the turbine engine.

16

The rotor blade system of any preceding clause, wherein the circumferential sweep is from greater than zero percent to forty percent of the span length.

The rotor blade system of any preceding clause, wherein the circumferential sweep is from five percent to twenty-five percent of the span length.

The rotor blade system of any preceding clause, wherein the circumferential sweep is from two percent to ten percent of the pitch.

The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, the base blade includes a base blade tip and a base blade root, and the blade tip and the blade root are coplanar with the base blade tip and the base blade root, respectively.

The rotor blade system of any preceding clause, wherein there is no sweep at the blade root.

The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a blade thickness between the leading edge and the trailing edge, the base blade defines a base blade thickness length between a leading edge of the base blade and a trailing edge of the base blade, and the blade thickness is within twenty-five percent of the base blade thickness.

The rotor blade system of the preceding clause, wherein the blade thickness is equal to the base blade thickness.

The rotor blade system of any preceding clause, further comprising a sweep point on the leading edge, wherein the leading edge axial sweep occurs from the sweep point to the blade tip and wherein the leading edge of the at least one rotor blade and a leading edge of the base blade are collinear from the sweep point to the blade root.

The rotor blade system of any preceding clause, further comprising a sweep point on the trailing edge, wherein the trailing edge axial sweep occurs from the sweep point to the blade tip and wherein the trailing edge of the at least one rotor blade and a trailing edge of the base blade are collinear from the sweep point to the blade root.

The rotor blade system of any preceding clause, further comprising a radial distance extending from a blade tip to a sweep point defining a point at which the leading edge axial sweep ceases and the leading edge of the at least one rotor blade is collinear with a leading edge of the base blade.

The rotor blade system of the preceding clause, wherein the radial distance is from greater than zero percent to forty percent of the span length of the at least one rotor blade.

The rotor blade system of the preceding clause, wherein the radial distance is five percent to twenty five percent of the span length of the last one rotor blade.

A rotor blade system for a turbine engine having a longitudinal centerline axis includes a rotor configured to rotate about the longitudinal centerline axis, and a plurality of rotor blades coupled circumferentially around the rotor, each rotor blade having a leading edge, a trailing edge, a suction side, and a pressure side. At least one rotor blade of the plurality of rotor blades has a suction side circumferential sweep of the suction side as compared to a suction side of a base blade of the plurality of rotor blades and a pressure side circumferential sweep of the pressure side as compared to a pressure side of the base blade of the plurality of rotor blades.

The rotor blade system of the preceding clause, wherein the suction side circumferential sweep is equal to the pressure side circumferential sweep.

The rotor blade system of any preceding clause, wherein the suction side circumferential sweep and the pressure side circumferential sweep extend in a direction of rotation of the turbine engine.

The rotor blade system of any preceding clause, wherein the suction side circumferential sweep and the pressure side circumferential sweep extend in an opposite direction of rotation of the turbine engine.

The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is an overlap between the blade tip and the base blade tip.

The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is no overlap between the blade tip and the base blade tip.

The rotor blade system of any preceding clause, wherein the base blade comprises a twist, and the at least one rotor blade comprises the same twist as that of the base blade.

The rotor blade system of any preceding clause, wherein the at least one rotor blade has no addition or removal of material or minimal addition or removal of material as compared to the base blade.

The rotor blade system of any preceding clause, wherein the at least one rotor blade comprises all of the plurality of rotor blades.

The rotor blade system of any preceding clause, wherein the at least one rotor blade comprises multiple rotor blades and the multiple rotor blades are arranged in a predetermined pattern about the rotor with a plurality of base blades.

The rotor blade system of any preceding clause, wherein the predetermined pattern is an alternating pattern.

The rotor blade system of any preceding clause, wherein the circumferential sweep is defined in an inter-blade spacing defined between the at least one rotor blade of the plurality of rotor blades and an adjacent rotor blade of the plurality of rotor blades.

The rotor blade system of any preceding clause, wherein a pitch is defined between a blade tip of adjacent rotor blades of the plurality of rotor blades, and the circumferential sweep is from greater than zero percent to twenty-five percent of the pitch.

The rotor blade system of any preceding clause, wherein the circumferential sweep is from two percent to ten percent of the pitch.

The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a span length between a blade tip and a blade root, and the suction side circumferential sweep or the pressure side circumferential sweep is from greater than zero percent to forty percent of the span length.

The rotor blade system of any preceding clause, wherein the suction side circumferential sweep or the pressure side circumferential sweep is from five percent to twenty-five percent of the span length.

The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a chord length between the leading edge and the trailing edge, the base blade defines a base blade chord length between a leading edge of the base blade and a trailing edge of the base blade, and the chord length is within ten percent of the base blade chord length.

The rotor blade system of any preceding clause, wherein the chord length is equal to the base blade chord length.

The rotor blade system of any preceding clause, further comprising an axial sweep, wherein the axial sweep is in a forward direction of the turbine engine.

The rotor blade system of any preceding clause, further comprising an axial sweep, wherein the axial sweep is in an aft direction of the turbine engine.

The rotor blade system of any preceding clause, further comprising an axial sweep, and wherein the axial sweep is in a forward direction of the turbine engine and the suction side circumferential sweep and the pressure side circumferential sweep are in a direction of rotation of the turbine engine.

The rotor blade system of any preceding clause, further comprising an axial sweep, and wherein the axial sweep is in a forward direction of the turbine engine and the suction side circumferential sweep, and the pressure side circumferential sweep are in a direction opposite of rotation of the turbine engine.

The rotor blade system of any preceding clause, further comprising an axial sweep, and wherein the axial sweep is in an aft direction of the turbine engine and the suction side circumferential sweep, and the pressure side circumferential sweep are in a direction of rotation of the turbine engine.

The rotor blade system of any preceding clause, further comprising an axial sweep, and wherein the axial sweep is in an aft direction of the turbine engine and the suction side circumferential sweep, and the pressure side circumferential sweep are in a direction opposite of rotation of the turbine engine.

The rotor blade system of any preceding clause, wherein the axial sweep is from greater than zero percent to forty percent of the span length.

The rotor blade system of any preceding clause, wherein the axial sweep is from five percent to twenty-five percent of the span length.

The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a chord length between the leading edge and the trailing edge, and wherein the axial sweep is from greater than zero percent to fifteen percent of the chord length.

The rotor blade system of any preceding clause, wherein the axial sweep is from five percent to ten percent of the chord length.

The rotor blade system of any preceding clause, wherein the axial sweep includes a leading edge axial sweep that is equal to a trailing edge axial sweep.

The rotor blade system of any preceding clause, wherein the leading edge axial sweep and the trailing edge axial sweep extend in a forward direction of the turbine engine.

The rotor blade system of any preceding clause, wherein the leading edge axial sweep and the trailing edge axial sweep extend in an aft direction of the turbine engine.

The rotor blade system of any preceding clause, wherein the at least one rotor blade includes a blade tip and a blade root, the base blade includes a base blade tip and a base blade root, and the blade tip and the blade root are coplanar with the base blade tip and the base blade root, respectively.

The rotor blade system of any preceding clause, wherein there is no sweep at the blade root.

The rotor blade system of any preceding clause, wherein the at least one rotor blade defines a blade thickness between the leading edge and the trailing edge, the base blade defines a base blade thickness length between a leading edge of the base blade and a trailing edge of the base blade, and the blade thickness is within twenty-five percent of the base blade thickness.

The rotor blade system of the preceding clause, wherein the blade thickness is equal to the base blade thickness.

The rotor blade system of any preceding clause, further comprising a sweep point on the suction side, wherein the suction side circumferential sweep occurs from the sweep point to the blade tip and wherein the suction side of the at

least one rotor blade and a suction side of the base blade are collinear from the sweep point to the blade root.

The rotor blade system of any preceding clause, further comprising a sweep point on the pressure side, wherein the pressure side circumferential sweep occurs from the sweep point to the blade tip and wherein the pressure side of the at least one rotor blade and a pressure side of the base blade are collinear from the sweep point to the blade root.

The rotor blade system of any preceding clause, further comprising a radial distance extending from a blade tip to a sweep point defining a point at which the suction side circumferential sweep ceases and the suction side of the at least one rotor blade is collinear with a suction side of the base blade.

The rotor blade system of the preceding clause, wherein the radial distance is from greater than zero percent to forty percent of the span length of the at least one rotor blade.

The rotor blade system of the preceding clause, wherein the radial distance is five percent to twenty five percent of the span length of the last one rotor blade.

A turbine engine comprising the rotor blade system of any preceding clause.

The turbine engine of the preceding clause, further comprising a low-pressure turbine, wherein the rotor blade system is included in at least one stage of the low-pressure turbine.

The turbine engine of any preceding clause, further comprising a high-pressure turbine, wherein the rotor blade system is included in at least one stage of the high-pressure turbine.

The turbine engine of any preceding clause, further comprising a low-pressure compressor, wherein the rotor blade system is included in at least one stage of the low-pressure compressor.

The turbine engine of any preceding clause, further comprising a high-pressure compressor, wherein the rotor blade system is included in at least one stage of the high-pressure compressor.

The turbine engine of any preceding clause, further comprising a fan, wherein the rotor blade system is included in the fan.

Although the foregoing description is directed to the preferred embodiments of the present disclosure, other variations and modifications will be apparent to those skilled in the art and may be made without departing from the spirit or the scope of the disclosure. Moreover, features described in connection with one embodiment of the present disclosure may be used in conjunction with other embodiments, even if not explicitly stated above.

The invention claimed is:

1. A rotor blade system for a turbine engine having a longitudinal centerline axis, the rotor blade system comprising:

a rotor configured to rotate about the longitudinal centerline axis; and

a plurality of rotor blades coupled circumferentially around the rotor, each rotor blade having a leading edge, a trailing edge, a suction side, and a pressure side, wherein at least one rotor blade of the plurality of rotor blades is a base blade, the base blade defining a base blade chord length between a leading edge of the base blade and a trailing edge of the base blade, and

wherein at least one rotor blade of the plurality of rotor blades is a swept blade, the swept blade having a leading edge axial sweep of the leading edge of the swept blade as compared to the leading edge of the base blade of the plurality of rotor blades and a trailing edge

axial sweep of the trailing edge of the swept blade as compared to the trailing edge of the base blade of the plurality of rotor blades and defining a swept blade chord length between the leading edge of the swept blade and the trailing edge of the swept blade,

wherein the swept blade chord length is a constant chord length with respect to the base blade chord length.

2. The rotor blade system of claim 1, wherein the leading edge axial sweep is equal to the trailing edge axial sweep.

3. The rotor blade system of claim 1, wherein the leading edge axial sweep and the trailing edge axial sweep extend in a forward direction of the turbine engine.

4. The rotor blade system of claim 1, wherein the leading edge axial sweep and the trailing edge axial sweep extend in an aft direction of the turbine engine.

5. The rotor blade system of claim 1, wherein the swept blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is an overlap between the blade tip and the base blade tip.

6. The rotor blade system of claim 1, wherein the swept blade includes a blade tip and a blade root, and the base blade includes a base blade tip and a base blade root, and there is no overlap between the blade tip and the base blade tip.

7. The rotor blade system of claim 1, wherein the base blade comprises a twist, and the swept blade comprises a same twist as that of the base blade.

8. The rotor blade system of claim 1, wherein the swept blade has no addition or removal of material or minimal addition or removal of material as compared to the base blade.

9. The rotor blade system of claim 1, wherein the swept blade comprises multiple rotor blades and the multiple rotor blades are arranged in a predetermined pattern about the rotor with a plurality of base blades.

10. The rotor blade system of claim 9, wherein the predetermined pattern is an alternating pattern.

11. The rotor blade system of claim 1, wherein the swept blade of the plurality of rotor blades further has a circumferential sweep in an inter-blade spacing defined between the swept blade of the plurality of rotor blades and an adjacent rotor blade of the plurality of rotor blades.

12. The rotor blade system of claim 11, wherein a pitch is defined between a blade tip of adjacent rotor blades of the plurality of rotor blades, and the circumferential sweep is greater than zero percent and less than or equal to twenty-five percent of the pitch.

13. The rotor blade system of claim 1, wherein the leading edge axial sweep or the trailing edge axial sweep is greater than zero percent and less than or equal to fifteen percent of the swept blade chord length.

14. The rotor blade system of claim 13, wherein the leading edge axial sweep or the trailing edge axial sweep is from five percent to ten percent of the swept blade chord length.

15. The rotor blade system of claim 1, wherein the swept blade defines a span length between a blade tip and a blade root, and the leading edge axial sweep or the trailing edge axial sweep is greater than zero percent and less than or equal to forty percent of the span length.

16. The rotor blade system of claim 15, wherein the leading edge axial sweep or the trailing edge axial sweep is from five percent to twenty-five percent of the span length.

17. The rotor blade system of claim 1, wherein the swept blade chord length is equal to the base blade chord length.

18. A turbine engine having a longitudinal centerline axis,
the turbine engine comprising:
a rotor blade system comprising:
a rotor configured to rotate about the longitudinal
centerline axis; and
a plurality of rotor blades coupled circumferentially
around the rotor, each rotor blade having a leading
edge, a trailing edge, a suction side, and a pressure
side, a low-pressure turbine, a high-pressure turbine,
a low-pressure compressor, and a high-pressure
compressor; and
a fan,
wherein the rotor blade system is included in at least one
stage of at least one of the low-pressure turbine, the
high-pressure turbine, the low-pressure compressor, the
high-pressure compressor, or the fan, and
wherein at least one rotor blade of the plurality of rotor
blades is a base blade, the base blade defining a base
blade chord length between a leading edge of the base
blade and a trailing edge of the base blade, and
wherein at least one rotor blade of the plurality of rotor
blades is a swept blade, the swept blade having a
leading edge axial sweep of the leading edge of the
swept blade as compared to the leading edge of the base
blade of the plurality of rotor blades and a trailing edge
axial sweep of the trailing edge of the swept blade as
compared to the trailing edge of the base blade of the
plurality of rotor blades and defining a swept blade
chord length between the leading edge of the swept
blade and the trailing edge of the swept blade,
wherein the swept blade chord length is a constant chord
length with respect to the base blade chord length.

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